

Mohammadreza Nemati

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EXPERIENCE

Senior AI Engineer

05/2025 - Present

Volkswagen Group of America

Belmont, CA

- Post-trained multimodal foundation models with SFT, DPO, and GRPO for specialized reasoner, grounder, verifier, and error-recovery components in real-device automotive workflows.
- Designed a long-horizon task planner that decomposed natural-language automotive human-machine interface (HMI) goals into executable steps and dynamically replanned from action feedback across multi-step device workflows.
- Built multimodal-agent evaluation infrastructure to benchmark task completion, integrating reasoning voice agents to accept user voice input and test in-vehicle HMI voice capabilities.
- Deployed and evaluated Qwen3.5 models on Azure ML managed endpoints with vLLM, configuring prefix caching, long-context inference, and throughput/latency monitoring.
- Compressed Qwen2.5-VL for autonomous-driving trajectory prediction from 3.75B to 1.4B using structured pruning, knowledge distillation from a 7B teacher, and quantization-aware training to reduce inference latency while maintaining accuracy under edge constraints.

Artificial Intelligence Research Assistant

08/2022 - 06/2025

Case Western Reserve University

Cleveland, Ohio

- Analyzed KV-cache key/value norm imbalance to optimize LLM inference memory, achieving 2× memory reduction with >94% accuracy retention across foundation models up to 70B parameters [\[paper link\]](#)
- Developed LIT-LVM, a latent-variable method for structured interaction modeling that outperformed elastic net, hierarchical lasso, and factorization-machine baselines across simulated and real-world datasets [\[TMLR 2025\]](#)
- Developed DEASurv, a multimodal predictive model combining entity-specific embeddings, attention, and contrastive learning for large-scale medical datasets

Software Engineer - AI/ML Intern

02/2024 - 07/2024

Sherwin-Williams

Cleveland, Ohio

- Built a VLM-based grounded-perception system integrating LLaVA and vision transformers for natural-language object grounding and segmentation.
- Designed and deployed real-time vision-transformer inference pipeline for SAM, tuning latency and improving segmentation quality by 7% IoU through inference evaluation and pipeline optimization.

Data Science Research Intern

05/2023 - 09/2023

Beyond Limits AI

Glendale, California

- Fine-tuned Llama-2 using PyTorch FSDP/DDP for scalable multi-GPU training and quantized deployment.
- Developed and deployed Graph Neural Network models (GCN, GAT, GraphSAGE) using PyTorch Geometric for graph-structured data, achieving 10% accuracy improvement over baseline approaches

SKILLS

Foundation Models / Multimodal: Transformers, Vision-Language Models, Multimodal Learning, Visual Grounding

Agentic AI: Long-Horizon Planning, Voice Agents, Tool Use, GUI Agents, Multi-Agent Workflows

Training / Evaluation: SFT, DPO, GRPO, Reinforcement Learning, PyTorch, Distributed Training (FSDP/DDP), Model and Agent Evaluation, Knowledge Distillation, Quantization, Pruning

Programming / Infrastructure: Python, C++, CUDA, vLLM, Docker, Git

EDUCATION

Case Western Reserve University – PhD, Computer Science

Cleveland, OH | 2025

University of Toledo – MS, Computer Science

Toledo, OH | 2020

Shiraz University – BS, Electrical Engineering

Shiraz, Iran | 2017